

Neuro-Symbolic Suffix Prediction with Stochastic Automata-Guided Decoding

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1 Supplementary Performance Results

Table S.1. RQ1 predictive accuracy (Normalized Damerau-Levenshtein similarity (nDLS)) results using procedural knowledge (mean $\pm$ standard deviation). Bold and underlined values denote the best and second-best distinct means per event log, respectively.

Event log	NN	DIFF-ERO _L	DIFF-ERO _G	NN $\otimes \mathcal{BK}$	NN $\rightarrow$ SDFA	SDFA _{NP}	SDFA
BPIC2012	<u>0.882$\pm$0.005</u>	0.722 $\pm$ 0.002	0.717 $\pm$ 0.010	<u>0.882$\pm$0.005</u>	0.886$\pm$0.000	0.886$\pm$0.000	0.886$\pm$0.000
BPIC2013-I	<u>0.831$\pm$0.028</u>	0.702 $\pm$ 0.004	0.703 $\pm$ 0.005	0.779 $\pm$ 0.107	0.854$\pm$0.000	0.854$\pm$0.000	0.854$\pm$0.000
BPIC2013-CP	0.845 $\pm$ 0.046	0.901 $\pm$ 0.008	0.895 $\pm$ 0.000	0.907 $\pm$ 0.004	<u>0.921$\pm$0.008</u>	0.928$\pm$0.000	0.928$\pm$0.000
Dreyers	0.372 $\pm$ 0.230	<u>0.551$\pm$0.018</u>	0.537 $\pm$ 0.034	0.487 $\pm$ 0.187	0.485 $\pm$ 0.191	0.748$\pm$0.006	0.748$\pm$0.006
Helpdesk	0.919 $\pm$ 0.012	0.922 $\pm$ 0.019	<u>0.935$\pm$0.000</u>	0.936$\pm$0.000	0.919 $\pm$ 0.012	0.936$\pm$0.000	0.936$\pm$0.000
Domestic	0.931 $\pm$ 0.005	0.893 $\pm$ 0.002	0.895 $\pm$ 0.000	0.940$\pm$0.001	0.940$\pm$0.001	<u>0.939$\pm$0.001</u>	<u>0.939$\pm$0.001</u>
International	0.944 $\pm$ 0.007	0.832 $\pm$ 0.036	0.767 $\pm$ 0.068	<u>0.959$\pm$0.008</u>	0.952 $\pm$ 0.002	0.968$\pm$0.000	0.968$\pm$0.000
Permit	0.658 $\pm$ 0.014	0.566 $\pm$ 0.051	0.459 $\pm$ 0.033	0.643 $\pm$ 0.042	<u>0.678$\pm$0.004</u>	0.695$\pm$0.000	0.695$\pm$0.000
Prepaid	0.947 $\pm$ 0.001	0.905 $\pm$ 0.003	0.916 $\pm$ 0.028	0.950$\pm$0.002	<u>0.948$\pm$0.000</u>	0.948$\pm$0.000	0.948$\pm$0.000
Request	<u>0.890$\pm$0.033</u>	0.886 $\pm$ 0.000	0.878 $\pm$ 0.011	0.929$\pm$0.000	0.929$\pm$0.000	0.929$\pm$0.000	0.929$\pm$0.000

Table S.2. RQ2 predictive compliance results using procedural knowledge (mean $\pm$ standard deviation). Bold and underlined values denote the best and second-best distinct means per event log, respectively.

Event log	NN	DIFF-ERO _L	DIFF-ERO _G	NN $\otimes \mathcal{BK}$	NN $\rightarrow$ SDFA	SDFA _{NP}	SDFA
(a) Feasibility $\uparrow$							
BPIC2012	1.000$\pm$0.000	0.926 $\pm$ 0.005	<u>0.930$\pm$0.005</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
BPIC2013-I	<u>0.973$\pm$0.039</u>	0.625 $\pm$ 0.023	0.626 $\pm$ 0.025	0.880 $\pm$ 0.170	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
BPIC2013-CP	1.000$\pm$0.000	0.954 $\pm$ 0.019	0.927 $\pm$ 0.000	0.960 $\pm$ 0.007	<u>0.987$\pm$0.019</u>	1.000$\pm$0.000	1.000$\pm$0.000
Dreyers	<u>0.802$\pm$0.006</u>	0.659 $\pm$ 0.000	0.708 $\pm$ 0.069	0.542 $\pm$ 0.324	0.448 $\pm$ 0.390	1.000$\pm$0.000	1.000$\pm$0.000
Helpdesk	1.000$\pm$0.000	<u>0.973$\pm$0.039</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Domestic	1.000$\pm$0.000	0.776 $\pm$ 0.014	<u>0.786$\pm$0.000</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
International	0.945 $\pm$ 0.014	0.772 $\pm$ 0.069	0.656 $\pm$ 0.156	0.941 $\pm$ 0.057	<u>0.959$\pm$0.000</u>	1.000$\pm$0.000	1.000$\pm$0.000
Permit	<u>0.998$\pm$0.003</u>	0.781 $\pm$ 0.044	0.404 $\pm$ 0.006	0.907 $\pm$ 0.038	0.974 $\pm$ 0.001	1.000$\pm$0.000	1.000$\pm$0.000
Prepaid	1.000$\pm$0.000	0.861 $\pm$ 0.003	0.899 $\pm$ 0.062	<u>0.994$\pm$0.008</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Request	<u>0.900$\pm$0.112</u>	0.788 $\pm$ 0.000	0.772 $\pm$ 0.023	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
(b) Termination $\uparrow$							
BPIC2012	1.000$\pm$0.000	0.796 $\pm$ 0.004	0.793 $\pm$ 0.013	<u>0.995$\pm$0.007</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
BPIC2013-I	<u>0.973$\pm$0.039</u>	0.625 $\pm$ 0.023	0.626 $\pm$ 0.025	0.880 $\pm$ 0.170	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
BPIC2013-CP	0.943 $\pm$ 0.050	0.949 $\pm$ 0.017	0.927 $\pm$ 0.000	0.958 $\pm$ 0.009	<u>0.985$\pm$0.018</u>	1.000$\pm$0.000	1.000$\pm$0.000
Dreyers	<u>0.802$\pm$0.006</u>	0.612 $\pm$ 0.000	0.612 $\pm$ 0.000	0.353 $\pm$ 0.457	0.448 $\pm$ 0.390	1.000$\pm$0.000	1.000$\pm$0.000
Helpdesk	1.000$\pm$0.000	<u>0.973$\pm$0.039</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Domestic	1.000$\pm$0.000	0.776 $\pm$ 0.014	<u>0.786$\pm$0.000</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
International	0.945 $\pm$ 0.014	0.768 $\pm$ 0.066	0.656 $\pm$ 0.156	0.941 $\pm$ 0.057	<u>0.959$\pm$0.000</u>	1.000$\pm$0.000	1.000$\pm$0.000
Permit	<u>0.996$\pm$0.001</u>	0.621 $\pm$ 0.271	0.319 $\pm$ 0.058	0.905 $\pm$ 0.039	0.970 $\pm$ 0.003	1.000$\pm$0.000	1.000$\pm$0.000
Prepaid	1.000$\pm$0.000	0.861 $\pm$ 0.003	0.899 $\pm$ 0.062	<u>0.994$\pm$0.008</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Request	<u>0.900$\pm$0.112</u>	0.788 $\pm$ 0.000	0.772 $\pm$ 0.023	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000

Table S.3. RQ3 decoding-efficiency results using procedural knowledge (mean $\pm$ standard deviation). Bold and underlined values denote the best and second-best distinct means per event log, respectively.

Event log	NN	DIFF-ERO _L	DIFF-ERO _G	NN $\otimes$ $\mathcal{B}\mathcal{K}$	NN $\rightarrow$ SDFA	SDFA _{NP}	SDFA
(a) Prediction time (s) $\downarrow$							
BPIC2012	1.429 $\pm$ 0.054	0.068$\pm$0.000	0.181 $\pm$ 0.064	2.433 $\pm$ 0.592	5.785 $\pm$ 0.596	2.364 $\pm$ 0.599	<u>0.088$\pm$0.002</u>
BPIC2013-I	1.014 $\pm$ 0.086	0.042$\pm$0.006	0.194 $\pm$ 0.092	2.015 $\pm$ 0.219	4.044 $\pm$ 0.285	1.675 $\pm$ 0.077	<u>0.116$\pm$0.016</u>
BPIC2013-CP	0.563 $\pm$ 0.029	0.017$\pm$0.000	<u>0.101$\pm$0.019</u>	1.775 $\pm$ 0.124	1.898 $\pm$ 0.106	1.769 $\pm$ 0.174	0.314 $\pm$ 0.006
Dreyers	31.742 $\pm$ 13.777	<u>0.883$\pm$0.117</u>	1.960 $\pm$ 0.256	123.348 $\pm$ 22.628	72.336 $\pm$ 4.677	97.928 $\pm$ 1.520	0.382$\pm$0.024
Helpdesk	1.335 $\pm$ 0.010	0.058$\pm$0.013	<u>0.143$\pm$0.027</u>	4.859 $\pm$ 0.069	2.601 $\pm$ 0.101	3.754 $\pm$ 0.367	0.384 $\pm$ 0.010
Domestic	4.583 $\pm$ 0.264	<u>0.205$\pm$0.045</u>	0.202$\pm$0.030	7.452 $\pm$ 0.425	12.596 $\pm$ 0.957	6.950 $\pm$ 0.243	0.309 $\pm$ 0.028
International	32.518 $\pm$ 1.023	<u>1.465$\pm$0.025</u>	2.836 $\pm$ 0.870	57.277 $\pm$ 0.740	67.542 $\pm$ 6.963	68.105 $\pm$ 26.217	0.799$\pm$0.107
Permit	45.329 $\pm$ 1.426	0.981$\pm$0.078	1.392 $\pm$ 0.719	114.732 $\pm$ 35.935	71.115 $\pm$ 5.080	114.607 $\pm$ 4.409	<u>1.275$\pm$0.068</u>
Prepaid	17.985 $\pm$ 0.907	0.623$\pm$0.055	1.892 $\pm$ 0.388	83.169 $\pm$ 27.220	78.930 $\pm$ 49.111	58.524 $\pm$ 1.809	<u>0.641$\pm$0.043</u>
Request	4.113 $\pm$ 0.266	0.169$\pm$0.001	0.358 $\pm$ 0.025	16.085 $\pm$ 1.496	8.630 $\pm$ 0.556	12.349 $\pm$ 3.479	<u>0.241$\pm$0.006</u>
(b) Normalized explored branch count (nEBC) $\downarrow$							
BPIC2012	0.182 $\pm$ 0.003	0.148 $\pm$ 0.002	<u>0.145$\pm$0.002</u>	0.183 $\pm$ 0.003	0.793 $\pm$ 0.023	0.188 $\pm$ 0.000	0.008$\pm$0.000
BPIC2013-I	0.222 $\pm$ 0.015	<u>0.191$\pm$0.006</u>	<u>0.191$\pm$0.005</u>	0.232 $\pm$ 0.006	0.751 $\pm$ 0.042	0.241 $\pm$ 0.000	0.018$\pm$0.000
BPIC2013-CP	<u>0.218$\pm$0.020</u>	<u>0.218$\pm$0.008</u>	0.228 $\pm$ 0.000	0.223 $\pm$ 0.009	0.726 $\pm$ 0.084	0.223 $\pm$ 0.000	0.041$\pm$0.000
Dreyers	0.251 $\pm$ 0.108	0.180 $\pm$ 0.016	<u>0.171$\pm$0.020</u>	0.331 $\pm$ 0.068	0.447 $\pm$ 0.015	0.330 $\pm$ 0.000	0.001$\pm$0.000
Helpdesk	0.410 $\pm$ 0.007	<u>0.367$\pm$0.008</u>	<u>0.372$\pm$0.000</u>	0.400 $\pm$ 0.000	0.680 $\pm$ 0.001	0.388 $\pm$ 0.000	0.042$\pm$0.000
Domestic	0.326 $\pm$ 0.015	0.298 $\pm$ 0.004	<u>0.290$\pm$0.004</u>	0.328 $\pm$ 0.010	0.632 $\pm$ 0.032	0.352 $\pm$ 0.010	0.006$\pm$0.000
International	0.354 $\pm$ 0.010	0.332 $\pm$ 0.007	<u>0.316$\pm$0.016</u>	0.362 $\pm$ 0.006	0.575 $\pm$ 0.023	0.375 $\pm$ 0.000	0.003$\pm$0.000
Permit	0.404 $\pm$ 0.003	0.220 $\pm$ 0.023	<u>0.202$\pm$0.015</u>	0.434 $\pm$ 0.018	0.650 $\pm$ 0.027	0.425 $\pm$ 0.000	0.006$\pm$0.000
Prepaid	0.323 $\pm$ 0.010	<u>0.286$\pm$0.005</u>	0.287 $\pm$ 0.018	0.328 $\pm$ 0.006	0.611 $\pm$ 0.032	0.334 $\pm$ 0.000	0.004$\pm$0.000
Request	0.309 $\pm$ 0.018	<u>0.298$\pm$0.000</u>	0.311 $\pm$ 0.016	0.319 $\pm$ 0.005	0.650 $\pm$ 0.037	0.337 $\pm$ 0.000	0.005$\pm$0.000

Table S.4. RQ1 predictive accuracy (Normalized Damerau-Levenshtein similarity (nDLS)) results using declarative knowledge (mean $\pm$ standard deviation). Bold and underlined values denote the best and second-best distinct means per event log, respectively.

Event log	NN	NN _{LLL}	NN _{GLL}	NN $\otimes$ $\mathcal{B}\mathcal{K}$	NN $\rightarrow$ SDFA	SDFA _{NP}	SDFA
BPIC2012	<u>0.882$\pm$0.005</u>	0.886$\pm$0.000	0.886$\pm$0.000	<u>0.882$\pm$0.005</u>	<u>0.882$\pm$0.005</u>	0.886$\pm$0.000	0.886$\pm$0.000
BPIC2013-I	0.831 $\pm$ 0.028	0.854$\pm$0.000	0.854$\pm$0.000	<u>0.844$\pm$0.010</u>	<u>0.844$\pm$0.009</u>	0.854$\pm$0.000	0.854$\pm$0.000
BPIC2013-CP	0.845 $\pm$ 0.046	0.882 $\pm$ 0.000	0.882 $\pm$ 0.000	0.910 $\pm$ 0.002	<u>0.911$\pm$0.001</u>	0.928$\pm$0.000	0.928$\pm$0.000
Dreyers	0.372 $\pm$ 0.230	<u>0.605$\pm$0.002</u>	0.559 $\pm$ 0.031	0.567 $\pm$ 0.054	0.285 $\pm$ 0.255	0.651$\pm$0.077	0.651$\pm$0.077
Helpdesk	0.919 $\pm$ 0.012	0.910 $\pm$ 0.000	0.910 $\pm$ 0.000	<u>0.927$\pm$0.012</u>	0.919 $\pm$ 0.012	0.936$\pm$0.000	0.936$\pm$0.000
Domestic	0.931 $\pm$ 0.005	<u>0.938$\pm$0.000</u>	<u>0.938$\pm$0.000</u>	0.931 $\pm$ 0.005	0.929 $\pm$ 0.006	0.941$\pm$0.000	0.941$\pm$0.000
International	0.944 $\pm$ 0.007	0.968$\pm$0.000	<u>0.965$\pm$0.001</u>	0.952 $\pm$ 0.007	0.954 $\pm$ 0.005	0.968$\pm$0.000	0.968$\pm$0.000
Permit	0.658 $\pm$ 0.014	0.696$\pm$0.000	0.696$\pm$0.000	0.662 $\pm$ 0.009	<u>0.663$\pm$0.009</u>	0.696$\pm$0.000	0.696$\pm$0.000
Prepaid	0.947 $\pm$ 0.001	<u>0.948$\pm$0.000</u>	<u>0.948$\pm$0.000</u>	0.946 $\pm$ 0.002	0.947 $\pm$ 0.001	0.956$\pm$0.000	0.956$\pm$0.000
Request	0.890 $\pm$ 0.033	0.929$\pm$0.000	0.929$\pm$0.000	<u>0.917$\pm$0.016</u>	0.913 $\pm$ 0.017	0.929$\pm$0.000	0.929$\pm$0.000

Table S.5. RQ2 predictive compliance results using declarative knowledge (mean $\pm$ standard deviation). Bold and underlined values denote the best and second-best distinct means per event log, respectively.

Event log	NN	NN _{LLL}	NN _{GLL}	NN $\otimes \mathcal{BK}$	NN $\rightarrow$ SDFA	SDFA _{NP}	SDFA
(a) Feasibility $\uparrow$							
BPIC2012	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
BPIC2013-I	<u>0.973$\pm$0.039</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
BPIC2013-CP	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Dreyers	0.802 $\pm$ 0.006	1.000$\pm$0.000	1.000$\pm$0.000	<u>0.913$\pm$0.097</u>	0.871 $\pm$ 0.091	1.000$\pm$0.000	1.000$\pm$0.000
Helpdesk	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Domestic	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
International	0.945 $\pm$ 0.014	1.000$\pm$0.000	<u>0.994$\pm$0.000</u>	0.971 $\pm$ 0.015	0.975 $\pm$ 0.012	1.000$\pm$0.000	1.000$\pm$0.000
Permit	<u>0.998$\pm$0.003</u>	1.000$\pm$0.000	1.000$\pm$0.000	<u>0.998$\pm$0.003</u>	<u>0.998$\pm$0.003</u>	1.000$\pm$0.000	1.000$\pm$0.000
Prepaid	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Request	0.900 $\pm$ 0.112	1.000$\pm$0.000	1.000$\pm$0.000	<u>0.985$\pm$0.021</u>	<u>0.985$\pm$0.021</u>	1.000$\pm$0.000	1.000$\pm$0.000
(b) Termination $\uparrow$							
BPIC2012	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
BPIC2013-I	<u>0.973$\pm$0.039</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
BPIC2013-CP	<u>0.943$\pm$0.050</u>	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Dreyers	0.802 $\pm$ 0.006	1.000$\pm$0.000	1.000$\pm$0.000	<u>0.913$\pm$0.097</u>	0.871 $\pm$ 0.091	1.000$\pm$0.000	1.000$\pm$0.000
Helpdesk	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Domestic	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
International	0.945 $\pm$ 0.014	1.000$\pm$0.000	<u>0.994$\pm$0.000</u>	0.971 $\pm$ 0.015	0.975 $\pm$ 0.012	1.000$\pm$0.000	1.000$\pm$0.000
Permit	0.996 $\pm$ 0.001	1.000$\pm$0.000	1.000$\pm$0.000	<u>0.998$\pm$0.003</u>	<u>0.998$\pm$0.003</u>	1.000$\pm$0.000	1.000$\pm$0.000
Prepaid	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000	1.000$\pm$0.000
Request	0.900 $\pm$ 0.112	1.000$\pm$0.000	1.000$\pm$0.000	<u>0.985$\pm$0.021</u>	<u>0.985$\pm$0.021</u>	1.000$\pm$0.000	1.000$\pm$0.000

Table S.6. RQ3 decoding-efficiency results using declarative knowledge (mean $\pm$ standard deviation). Bold and underlined values denote the best and second-best distinct means per event log, respectively.

Event log	NN	NN _{LLL}	NN _{GLL}	NN $\otimes \mathcal{BK}$	NN $\rightarrow$ SDFA	SDFA _{NP}	SDFA
(a) Prediction time (s) $\downarrow$							
BPIC2012	1.429 $\pm$ 0.054	0.102 $\pm$ 0.026	<u>0.089$\pm$0.010</u>	1.712 $\pm$ 0.180	5.830 $\pm$ 0.171	0.090 $\pm$ 0.011	0.020$\pm$0.002
BPIC2013-I	1.014 $\pm$ 0.086	0.058 $\pm$ 0.000	0.062 $\pm$ 0.001	1.782 $\pm$ 0.025	3.103 $\pm$ 0.131	<u>0.051$\pm$0.002</u>	0.023$\pm$0.002
BPIC2013-CP	0.563 $\pm$ 0.029	0.034 $\pm$ 0.000	0.036 $\pm$ 0.005	0.901 $\pm$ 0.050	1.494 $\pm$ 0.122	<u>0.025$\pm$0.001</u>	0.014$\pm$0.000
Dreyers	31.742 $\pm$ 13.777	2.299 $\pm$ 0.010	1.998 $\pm$ 0.027	7.556 $\pm$ 0.142	52.429 $\pm$ 3.007	<u>1.995$\pm$0.099</u>	1.283$\pm$0.052
Helpdesk	1.335 $\pm$ 0.010	0.077 $\pm$ 0.000	0.089 $\pm$ 0.014	2.150 $\pm$ 0.018	1.992 $\pm$ 0.055	<u>0.065$\pm$0.000</u>	0.013$\pm$0.000
Domestic	4.583 $\pm$ 0.264	0.315 $\pm$ 0.035	0.227 $\pm$ 0.004	7.652 $\pm$ 0.857	10.478 $\pm$ 0.885	<u>0.205$\pm$0.003</u>	0.152$\pm$0.005
International	32.518 $\pm$ 1.023	<u>2.024$\pm$0.099</u>	2.466 $\pm$ 0.147	34.786 $\pm$ 0.854	57.190 $\pm$ 2.287	3.145 $\pm$ 1.764	0.204$\pm$0.001
Permit	45.329 $\pm$ 1.426	3.044 $\pm$ 0.223	2.378 $\pm$ 0.010	64.825 $\pm$ 8.884	60.152 $\pm$ 1.324	<u>2.337$\pm$0.032</u>	0.771$\pm$0.018
Prepaid	17.985 $\pm$ 0.907	<u>0.991$\pm$0.029</u>	1.064 $\pm$ 0.010	19.443 $\pm$ 1.138	34.584 $\pm$ 3.031	0.998 $\pm$ 0.048	0.173$\pm$0.013
Request	4.113 $\pm$ 0.266	0.229 $\pm$ 0.003	0.239 $\pm$ 0.010	4.553 $\pm$ 0.196	8.872 $\pm$ 1.094	<u>0.205$\pm$0.011</u>	0.029$\pm$0.001
(b) Normalized explored branch count (nEBC) $\downarrow$							
BPIC2012	0.182 $\pm$ 0.003	0.196 $\pm$ 0.006	0.197 $\pm$ 0.006	0.181 $\pm$ 0.001	0.793 $\pm$ 0.023	<u>0.180$\pm$0.001</u>	0.035$\pm$0.000
BPIC2013-I	<u>0.222$\pm$0.015</u>	0.263 $\pm$ 0.001	0.264 $\pm$ 0.000	0.234 $\pm$ 0.006	0.751 $\pm$ 0.042	0.231 $\pm$ 0.000	0.085$\pm$0.001
BPIC2013-CP	<u>0.218$\pm$0.020</u>	0.288 $\pm$ 0.001	0.284 $\pm$ 0.002	0.225 $\pm$ 0.006	0.726 $\pm$ 0.084	0.222 $\pm$ 0.000	0.110$\pm$0.000
Dreyers	<u>0.251$\pm$0.108</u>	0.299 $\pm$ 0.001	0.287 $\pm$ 0.007	0.527 $\pm$ 0.000	0.447 $\pm$ 0.015	0.283 $\pm$ 0.016	0.185$\pm$0.012
Helpdesk	0.410 $\pm$ 0.007	0.441 $\pm$ 0.003	0.425 $\pm$ 0.011	0.405 $\pm$ 0.008	0.680 $\pm$ 0.001	<u>0.383$\pm$0.000</u>	0.068$\pm$0.000
Domestic	0.326 $\pm$ 0.015	0.354 $\pm$ 0.002	0.353 $\pm$ 0.005	0.326 $\pm$ 0.015	0.632 $\pm$ 0.032	<u>0.317$\pm$0.004</u>	0.236$\pm$0.003
International	<u>0.354$\pm$0.010</u>	0.373 $\pm$ 0.000	0.361 $\pm$ 0.000	0.374 $\pm$ 0.005	0.575 $\pm$ 0.023	0.362 $\pm$ 0.002	0.038$\pm$0.000
Permit	0.404 $\pm$ 0.003	0.414 $\pm$ 0.001	0.408 $\pm$ 0.001	0.400 $\pm$ 0.003	0.650 $\pm$ 0.027	<u>0.394$\pm$0.006</u>	0.130$\pm$0.002
Prepaid	0.323 $\pm$ 0.010	0.327 $\pm$ 0.001	0.323 $\pm$ 0.004	0.332 $\pm$ 0.005	0.611 $\pm$ 0.032	<u>0.320$\pm$0.000</u>	0.054$\pm$0.000
Request	0.309 $\pm$ 0.018	0.341 $\pm$ 0.001	0.331 $\pm$ 0.001	0.342 $\pm$ 0.010	0.650 $\pm$ 0.037	<u>0.302$\pm$0.001</u>	0.041$\pm$0.001